



Collocation method for stochastic delay differential equations

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ABSTRACT

In this work, we present a collocation-based numerical approach for handling stochastic delay differential equations. We approximate the solution function, and after that, we carry out integrations between the predefined collocation points to achieve a mapping from the delayed state to the present state. We build the first and second moment mapping matrices based on the mapping, and we utilize the matrices to approximate the stationary first and second moments and their stability. Numerical studies of a first and second-order stochastic delay differential equation show the convergence and time complexity of the stochastic collocation method. The last section covers the issues and possible further improvements of the method.

1. Introduction

Delay differential equations are widely used to describe and investigate systems and processes where time delay plays an important role, such as delay control problems [1], phantom traffic jams [2], machine tool vibrations [3,4] or human balancing [5]. In many cases, these systems are inherently subjected to uncertainty, to random fluctuations and to unpredictable disturbances. Such random effects are the noisy environment in case of control problems, random parameter fluctuations in case of traffic models including human interactions or the high-frequency stochastic variation of the cutting force [6] in case of machining processes.

Stochastic delay differential equations (SDDEs) are powerful mathematical modelling and simulation tools for these systems and processes, as they provide a versatile platform for investigating problems that include both stochastic effects and time delay. However, investigating the stability and steady-state behaviour of SDDEs present a number of challenges, such as the lack of analytical solutions or the scarcity of efficient numerical methods specialized for SDDEs.

Furthermore, there are several definitions for the stability properties of linear stochastic systems. One definition is the almost sure exponential stability [7–9], which investigates the mean decay rate of the exponential envelope of a path-wise solution of the stochastic system. This approach allows a finite number of bursts in the trajectories, meaning that the system might not even have a steady-state solution: such behaviour prevents the effective use in engineering applications. Such a system appears in the model of regenerative machine tool vibrations, where the cutting tool vibrates due to the stochastic force

fluctuations, or in the model of delayed feedback control where the controlled system is excited by the noisy environment. If these systems have the aforementioned bursts, the models predict unacceptable operational conditions. A more conservative stability definition is the so-called mean square exponential stability or second moment stability [10]. Here we investigate the stability of the mean evolution of the ensemble of the trajectories of the squared process, ensuring that the stable system has a steady-state solution. This is especially useful in engineering problems: if the system has a steady-state solution, we can also prescribe an envelope that restricts the fluctuations of the system for a given application.

There are a small number of methods to investigate the stability and steady-state behaviour of SDDEs. We can apply stochastic calculus for some simple SDDEs with constant coefficients to analyse the moment stability [11]. There are more general approaches, but these are usually problem-specific and only relevant to small degree-of-freedom systems. For example, we can transform the original SDDE into a non-delayed stochastic differential equation (SDE) in case of small delays [12,13]. Moreover, it is possible to check the fulfilment of the necessary and sufficient conditions for exponential and second moment stability by applying Lyapunov functionals [14,15]. Also stochastic averaging is a technique to study the long-term behaviour of dynamical systems, subjected to random disturbances [16]: we average the response of the system numerically or analytically to reduce the effect of noise and reveal its underlying behaviour.

Another approach is based on full discretization [17], where the mean square of a time step with the Explicit Euler–Maruyama scheme

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[18] is formulated as a matrix–vector multiplication. Then the mean square stability of the investigated system is approximated with the help of the spectral radius of the stepping matrix of the squared process. This approach produces deterministic results for the mean square stability, but the method is unreliable due to its extremely slow error convergence [19]. The stochastic semidiscretization method [19] builds on the semidiscretization approach [20], and significantly improves the error convergence of the stability and steady-state behaviour computations by using a high-order approximation of the delayed dynamics.

This work aims to introduce an algorithmically efficient collocation based numerical method to determine the mean square stability and stationary second moment of stochastic dynamical systems with delays. The collocation methods are already proven to be remarkably efficient regarding the stability analysis of deterministic delay differential equations [21–25] as it has an exponential error convergence rate [26,27]. Collocation is also applied in some stochastic approaches. A stochastic finite element method based on collocation is presented in [28], where the technique is utilized for a large class of mechanical problems, i.e. static, dynamic and spectral problems. Another approach uses collocation for derivative estimation [29]. The efficacy of stochastic collocation is presented in [30], and an investigation of the limits of stochastic collocation methods in solving partial differential equations through finite element applications is carried out in [31].

In this work, we focus on the steady-state second moment and the *second moment stability* of SDDEs, because this stability definition provides a strict condition for the behaviour of the fluctuations of the stochastic process. We use a weak formulation of the collocation method for SDDEs: we approximate the method with a linear combination of base functions, and construct an equation system for the coefficients of the base functions by formally evaluating the approximated solution at given collocation points. This approach incorporates the evolution of the Wiener process [32], which is used in this work to model the noise in the system. We use numerical experiments to investigate the error convergence of the collocation method and the computational time necessary to obtain the stability and the steady-state of the second moment process. To acquire reference solutions for the analysis, we use the stochastic semidiscretization method, as it produces similar metrics for the second moment stability condition as the introduced collocation-based approach.

In Section 2 we show the representation of stochastic delay differential equations using an approximation of the solution and we explain how to use the weak formulation of collocation to build the mapping equation between the present and the delayed states. In Section 3 we derive the first and second moment mapping equations and approximate the stationary moments and their stability. In Section 4 we investigate the error convergence properties and the CPU time complexity of the stochastic collocation method through two numerical case studies: the scalar stochastic Hayes equation and a second order system commonly used to model delayed feedback control. We also display stability charts and the corresponding steady-state second moment behaviour for these two examples. In Section 5 we summarize our work and findings and give some remarks on the possible improvements and issues of the method.

2. Collocation based mapping for stochastic delay differential equations

We consider linear time-invariant stochastic delay differential equations in the following incremental form

$$\begin{aligned} dx(t) &= (Ax(t) + Bx(t - \tau) + c)dt + (\beta x(t - \tau) + \gamma)dW(t), \\ x(t) &= \phi(t), \quad t \in [-\tau, 0]. \end{aligned} \quad (1)$$

Here $x(t)$ is the time t dependent $\mathbb{R}^d$ -valued state, $A, B, \beta \in \mathbb{R}^{d \times d}$ are the constant coefficient matrices, $c, \gamma \in \mathbb{R}^d$ are the additive vectors, τ

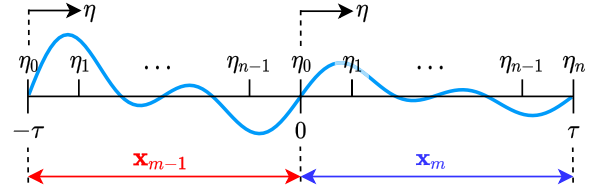


Fig. 1. The collocation points η_i and the solution segments x_{m-1} , x_m corresponding to the delayed and the present states, respectively, displayed on the $t \in [-\tau, \tau]$ time interval.

is the time delay. The noise is introduced to the system through the $\mathbb{R}$ -valued standard Wiener process $W(t)$ which has Gaussian distributed increments $dW(t) \sim \mathcal{N}(0, dt)$. We consider systems where the noise excitation has an additive and a multiplicative term which is proportional to the delayed state $x(t - \tau)$ only. We assume that the initial state $\phi(t) : [-\tau, 0] \rightarrow \mathbb{R}$ is continuous with $\mathbb{E}\|\phi\|_{L^\infty}^2 < \infty$ and $\phi(t)$ is F_0 measurable. However, these assumptions are just to ensure uniqueness and existence of path-wise realizations of x_t , ϕ_t is not necessary during the stability analysis.

In the following sections we transform the SDDE (1) to a stochastic mapping. The stochastic mapping is used to obtain the first and second moment mappings, which serve as the basis of the steady-state and stability analysis. The transformation has four key steps. First, we *approximate the solution segments* (present and history) by a linear combination of $n + 1$ basis functions. Second, we derive a *linear equation system* for the coefficients of the approximation. The third step is *using the first and second moment* of the equation system with respect to the coefficients of the approximation. Here, we obtain the first and second moment mappings between the approximated present and history segments. Finally, we *use the fixed points of the mappings* to obtain the steady-state moments, and the mapping matrices to analyse stability.

To prepare the mapping between the present and the delayed states, we approximate the SDDE (1) piecewise using a collocation based approach. We approximate the τ long solution segment $x(t_m)$, $t_m \in [(m-1)\tau, m\tau]$, with

$$x_m(t_m) := \sum_{i=0}^n \varphi_i(\eta_m) \mathbf{a}_{m,i}. \quad (2)$$

The approximation $x_m(t_m)$ is a linear combination of a set of basis functions $\varphi_i(\eta)$, $i = 0, 1, \dots, n$ with respect to the coefficient vectors $\mathbf{a}_{m,i}$. The coefficient vectors $\mathbf{a}_{m,i}$ are the unknowns of the system, because the history $x(t_{m-1})$ is given on the interval $t_{m-1} \in [(m-2)\tau, (m-1)\tau]$ for which $\mathbf{a}_{m-1,i}$ can be determined based on the history $\phi(t)$.

The basis functions $\varphi_i(\eta_m)$ have their local normalized time η_m . In this study Chebyshev functions and Chebyshev points are used as basis functions $\varphi_i(\eta_m)$ and collocation points $\eta_{m,i}$ [33]:

$$\varphi_i(\eta_m) = \cos(i \arccos(\eta_m)) \quad \text{and} \quad \eta_{m,i} = \cos\left(\frac{(2i+1)\pi}{2n}\right), \quad i = 0, 1, \dots, n, \quad (3)$$

and we use the linear transformation between t_m and η_m :

$$t_m(\eta_m) = \frac{\tau}{2}(\eta_m - 1) + \tau m \quad \text{and} \quad \eta_m(t_m) = \frac{2(t_m - \tau m)}{\tau} + 1. \quad (4)$$

Fig. 1 shows the collocation points along the solution segments. Note, that $\eta_m \in [-1, 1]$ does not depend on m , therefore, we omit the index m : $t_m(\eta) := t_m(\eta_m)$.

Next, we build the equation system for $\mathbf{a}_{m,i}$ and $\mathbf{a}_{m-1,i}$ by evaluating the approximation of the present state at the $n+1$ predefined collocation points $\eta_i = \eta_{m,i}$ in (3) [21]. First, we substitute $x_m(t_m)$ into (1) and then we integrate it between the discrete points η_{i-1} and η_i yielding

$$\begin{aligned} x_m(t_m(\eta_i)) &= x_m(t_m(\eta_{i-1})) + \int_{\eta_{i-1}}^{\eta_i} [Ax_m(t_m(\eta)) + Bx_{m-1}(t_{m-1}(\eta)) + c] \delta \eta \\ &\quad + \int_{\eta_{i-1}}^{\eta_i} [\beta x_{m-1}(t_{m-1}(\eta)) + \gamma] \sqrt{\delta \eta} dW_\eta. \end{aligned} \quad (5)$$

Here we used $dt = \delta d\eta$, where the transformation function $t_m(\eta)$ (4) determines the scaling factor $\delta = \tau/2$. Moreover, we use the substitution $dW(t) = \sqrt{\delta}dW(\eta)$ as well, to ensure that the variance of the rescaled Wiener increment $dW(\eta)$ is the same as the variance of $dW(t)$, namely $\mathbb{E}[dW(t)^2] = dt = \sqrt{\delta}d\eta = \mathbb{E}[\sqrt{\delta}dW(\eta)^2]$ holds.

Substituting (2) into the integral equation (5) yields

$$\begin{aligned} \sum_{i=0}^n \varphi_i(\eta_I) \mathbf{a}_{m,i} &= \sum_{i=0}^n \varphi_i(\eta_{I-1}) \mathbf{a}_{m,i} \\ &+ \int_{\eta_{I-1}}^{\eta_I} \left[\mathbf{A} \sum_{i=0}^n \varphi_i(\eta) \mathbf{a}_{m,i} + \mathbf{B} \sum_{i=0}^n \varphi_i(\eta) \mathbf{a}_{m-1,i} + \mathbf{c} \right] \delta d\eta \\ &+ \int_{\eta_{I-1}}^{\eta_I} \left[\beta \sum_{i=0}^n \varphi_i(\eta) \mathbf{a}_{m-1,i} + \gamma \right] \sqrt{\delta} dW_\eta, \quad I = 1, 2, \dots, n. \end{aligned} \quad (6)$$

The $(n+1)$ th equation describes the continuity condition between the solution segments $\mathbf{x}_m$ and $\mathbf{x}_{m-1}$, i.e.,

$$\begin{aligned} \mathbf{x}_m(0) &= \mathbf{x}_{m-1}(\tau), \\ \mathbf{x}_m(t_m(\eta_0)) &= \mathbf{x}_{m-1}(t_{m-1}(\eta_n)), \\ \sum_{i=0}^n \varphi_i(\eta_0) \mathbf{a}_{m,i} &= \sum_{i=0}^n \varphi_i(\eta_n) \mathbf{a}_{m-1,i}. \end{aligned} \quad (7)$$

Based on Eqs. (6) and (7) we build a matrix equation which describes the relation between the present and delayed states. The coefficients $\mathbf{a}_{m,i}$ and $\mathbf{a}_{m-1,i}$ are constants and independent of the local coordinate η therefore, they are separated from the integral. Collecting the integrated coefficients leads to:

$$\mathbf{E} \mathbf{a}_m = \mathbf{S} \mathbf{a}_{m-1} + \mathbf{C} \mathbf{a}_{m-1} + \mathbf{A}_\varphi \mathbf{a}_m + \mathbf{B}_\varphi \mathbf{a}_{m-1} + \mathbf{f} + \beta_\varphi \mathbf{a}_{m-1} + \mathbf{g}. \quad (8)$$

Here, the coefficient matrices $\mathbf{E}$, $\mathbf{S}$ and $\mathbf{C} \in \mathbb{R}^{(n+1)d \times (n+1)d}$ correspond to the present states, the starting points of the integrations and the connection equation, respectively. We use the integrals of the basis functions and the deterministic coefficient matrices of the original differential equation $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{c}$ to construct the matrices $\mathbf{A}_\varphi$, $\mathbf{B}_\varphi \in \mathbb{R}^{(n+1)d \times (n+1)d}$ and $\mathbf{f} \in \mathbb{R}^{(n+1)d}$, respectively. The stochastic terms appear in matrices β_φ and $\mathbf{g}$ in the form of stochastic integrals based on the stochastic coefficient matrices β and γ . The structure of the coefficients of (8) are listed in Eq. (9) in Box I where $\mathbf{I}_d$ is the d dimensional identity matrix and $\otimes$ denotes the Kronecker-product of two arbitrary matrices $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$:

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} A_{11}\mathbf{B} & \dots & A_{1n}\mathbf{B} \\ \vdots & & \vdots \\ A_{m1}\mathbf{B} & \dots & A_{mn}\mathbf{B} \end{bmatrix}_{(m+p) \times (n+q)}. \quad (10)$$

To obtain a more concise form of (8) we collect the coefficient matrices of $\mathbf{a}_m$ and $\mathbf{a}_{m-1}$ as

$$\mathbf{F} \mathbf{a}_m = (\mathbf{F}_\tau + \mathbf{G}_\tau) \mathbf{a}_{m-1} + (\mathbf{f} + \mathbf{g}), \quad (11)$$

where $\mathbf{F} = \mathbf{E} - \mathbf{S} - \mathbf{A}_\varphi$, $\mathbf{F}_\tau = \mathbf{C} + \mathbf{B}_\varphi$ and $\mathbf{G}_\tau = \beta_\varphi$. Note that the matrices $\mathbf{F}$, $\mathbf{F}_\tau$ and $\mathbf{f}$ are deterministic, while $\mathbf{G}_\tau$ and $\mathbf{g}$ are stochastic. We use the mapping (11) to obtain the approximate path-wise solution segments $\mathbf{x}_m$, considering that the coefficients $\mathbf{a}_0$ of the approximation of the initial function ϕ are available. We also need to approximate the Wiener integrals necessary to construct $\mathbf{G}_\tau$ and $\mathbf{g}$. In the following section we use (11) to construct the first and second moment mappings, that we use to analyse the steady-state behaviour of the first and second moment evolution of $\mathbf{x}$.

3. Moment mappings and steady-state behaviour

In this section we consider the evolution and steady-state behaviour of the first moment $\mathbb{E}[\mathbf{x}(t)]$ and second moment $\mathbb{E}[\mathbf{x}(t)\mathbf{x}(t)^T]$. We approximate the first and second moment evolution of $\mathbf{x}$ using the first and second moment of the mapping (11). In the proposed stochastic collocation method, the coefficients of the approximated state represent

these quantities. To determine the first and second moment mappings, first we consider the properties of the first and second moment of the Ito integrals [34], namely

$$\mathbb{E} \left[\int_{\eta_l}^{\eta_J} \varphi_i(\eta) dW_\eta \right] = 0, \quad (12)$$

$$\begin{aligned} \mathbb{E} \left[\int_{\eta_k}^{\eta_{k+1}} \varphi_i(\eta) dW_\eta \cdot \int_{\eta_l}^{\eta_{l+1}} \varphi_j(\eta) dW_\eta \right] \\ = \begin{cases} \int_{\eta_k}^{\eta_{k+1}} \varphi_i(\eta) \varphi_j(\eta) d\eta & \text{if } k = l, \\ 0 & \text{otherwise.} \end{cases} \end{aligned} \quad (13)$$

The property (13) is also known as Ito isometry.

3.1. First moment

Here, the expected value of the approximation coefficients $\bar{\mathbf{a}}_m := \mathbb{E}[\mathbf{a}_m]$ represent the first moment dynamics of the system defined in (1). To determine the first moment mapping we consider the expected value of (11), i.e.,

$$\mathbb{E}[\mathbf{F} \mathbf{a}_m] = \mathbb{E}[(\mathbf{F}_\tau + \mathbf{G}_\tau) \mathbf{a}_{m-1}] + \mathbb{E}[\mathbf{f}] + \mathbb{E}[\mathbf{g}]. \quad (14)$$

We utilize, that the elements of the matrix $\mathbf{G}_\tau$ are independent of the approximation coefficients $\mathbf{a}_{m-1}$ representing the delay state, since the integrals in $\mathbf{G}_\tau$ correspond to the time interval $t \in [(m-1)\tau, m\tau]$ that does not overlap with the delay state's time interval $t \in [(m-2)\tau, (m-1)\tau]$. This means, that we take the expected value of $\mathbf{G}_\tau$ and $\mathbf{a}_{m-1}$ separately. Moreover, we omit the expected value from the deterministic terms $\mathbf{F}$, $\mathbf{F}_\tau$ and $\mathbf{f}$ and use the property (12) in cases of the matrix $\mathbb{E}[\mathbf{G}_\tau] = \mathbf{0}$ and the vector $\mathbb{E}[\mathbf{g}] = \mathbf{0}$ leading to

$$\mathbf{F} \bar{\mathbf{a}}_m = \mathbf{F}_\tau \bar{\mathbf{a}}_{m-1} + \mathbf{f}. \quad (15)$$

Here, all the stochastic effects are averaged out. This essentially means that in this linear case the expected value of the stochastic system's solution is exactly the solution of the corresponding deterministic system [10,19,34] leading to the traditional (deterministic) collocation method.

We find the fixed-point of the mapping (15) by substituting $\bar{\mathbf{a}}_m$ and $\bar{\mathbf{a}}_{m-1}$ with the first moment of the stationary approximation coefficient $\bar{\mathbf{a}}_{\text{st}} := \lim_{m \rightarrow \infty} \bar{\mathbf{a}}_m$ and then solving the resulting equation for $\bar{\mathbf{a}}_{\text{st}}$:

$$\bar{\mathbf{a}}_{\text{st}} = (\mathbf{F} - \mathbf{F}_\tau)^{-1} \mathbf{f}. \quad (16)$$

Next we determine the stability of the stationary first moment by perturbing the stationary approximation coefficients: $\bar{\mathbf{a}}_m = \bar{\mathbf{a}}_{\text{st}} + \bar{\mathbf{a}}_m^*$. After substituting into (15), we get:

$$\mathbf{F}(\bar{\mathbf{a}}_{\text{st}} + \bar{\mathbf{a}}_m^*) = \mathbf{F}_\tau(\bar{\mathbf{a}}_{\text{st}} + \bar{\mathbf{a}}_{m-1}^*) + \mathbf{f}. \quad (17)$$

We use (16) to eliminate $\bar{\mathbf{a}}_{\text{st}}$ from (17) and we obtain the first moment evolution of the perturbed approximation coefficients $\bar{\mathbf{a}}_m^*$, that is

$$\mathbf{F} \bar{\mathbf{a}}_m^* = \mathbf{F}_\tau \bar{\mathbf{a}}_{m-1}^*. \quad (18)$$

We use the first moment mapping matrix $\mathbf{H}_1$ to determine the stability of the stationary first moment solution, where

$$\mathbf{H}_1 = \mathbf{F}^{-1} \mathbf{F}_\tau. \quad (19)$$

The first moment is asymptotically stable if the spectral radius $\rho(\mathbf{H}_1) < 1$ where ρ is defined as

$$\rho(\mathbf{H}_1) = \max_{\lambda_i} \{ |\lambda_i|, \lambda \in \mathbb{C} : \det(\mathbf{H}_1 - \lambda \mathbf{I}) = 0 \}. \quad (20)$$

This concludes that all the eigenvalues of $\mathbf{H}_1$ are inside the unit circle of the complex plane, therefore the mapping $\bar{\mathbf{a}}_{m-1}^* \mapsto \bar{\mathbf{a}}_m^*$ is a contraction.

$$\begin{aligned}
\mathbf{E} &= \begin{bmatrix} \varphi_0(\eta_0) & \varphi_1(\eta_0) & \cdots & \varphi_n(\eta_0) \\ \varphi_0(\eta_1) & \varphi_1(\eta_1) & \cdots & \varphi_n(\eta_1) \\ \vdots & \vdots & & \vdots \\ \varphi_0(\eta_n) & \varphi_1(\eta_n) & \cdots & \varphi_n(\eta_n) \end{bmatrix} \otimes \mathbf{I}_d, \\
\mathbf{A}_\varphi &= \begin{bmatrix} 0 & \cdots & 0 \\ \int_{\eta_0}^{\eta_1} \varphi_0(\eta) d\eta & \cdots & \int_{\eta_0}^{\eta_1} \varphi_n(\eta) d\eta \\ \vdots & & \vdots \\ \int_{\eta_{n-1}}^{\eta_n} \varphi_0(\eta) d\eta & \cdots & \int_{\eta_{n-1}}^{\eta_n} \varphi_n(\eta) d\eta \end{bmatrix} \otimes \delta \mathbf{A}, \\
\mathbf{C} &= \begin{bmatrix} \varphi_0(\eta_n) & \cdots & \varphi_n(\eta_n) \\ 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{bmatrix} \otimes \mathbf{I}_d, \\
\boldsymbol{\beta}_\varphi &= \begin{bmatrix} 0 & \cdots & 0 \\ \int_{\eta_0}^{\eta_1} \varphi_0(\eta) dW_\eta & \cdots & \int_{\eta_0}^{\eta_1} \varphi_n(\eta) dW_\eta \\ \vdots & & \vdots \\ \int_{\eta_{n-1}}^{\eta_n} \varphi_0(\eta) dW_\eta & \cdots & \int_{\eta_{n-1}}^{\eta_n} \varphi_n(\eta) dW_\eta \end{bmatrix} \otimes \sqrt{\delta} \boldsymbol{\beta}, \\
\mathbf{a}_m &= \begin{bmatrix} \mathbf{a}_{m,0} \\ \mathbf{a}_{m,1} \\ \vdots \\ \mathbf{a}_{m,n} \end{bmatrix}_{(n+1)d}, \\
\mathbf{S} &= \begin{bmatrix} 0 & \cdots & 0 \\ \varphi_0(\eta_0) & \cdots & \varphi_n(\eta_0) \\ \vdots & & \vdots \\ \varphi_0(\eta_{n-1}) & \cdots & \varphi_n(\eta_{n-1}) \end{bmatrix} \otimes \mathbf{I}_d, \\
\mathbf{B}_\varphi &= \begin{bmatrix} 0 & \cdots & 0 \\ \int_{\eta_0}^{\eta_1} \varphi_0(\eta) d\eta & \cdots & \int_{\eta_0}^{\eta_1} \varphi_n(\eta) d\eta \\ \vdots & & \vdots \\ \int_{\eta_{n-1}}^{\eta_n} \varphi_0(\eta) d\eta & \cdots & \int_{\eta_{n-1}}^{\eta_n} \varphi_n(\eta) d\eta \end{bmatrix} \otimes \delta \mathbf{B}, \\
\mathbf{f} &= \begin{bmatrix} 0 \\ \int_{\eta_0}^{\eta_1} 1 d\eta \\ \vdots \\ \int_{\eta_{n-1}}^{\eta_n} 1 d\eta \end{bmatrix} \otimes \delta \mathbf{c}, \\
\mathbf{g} &= \begin{bmatrix} 0 \\ \int_{\eta_0}^{\eta_1} 1 dW_\eta \\ \vdots \\ \int_{\eta_{n-1}}^{\eta_n} 1 dW_\eta \end{bmatrix} \otimes \sqrt{\delta} \boldsymbol{\gamma}, \\
\mathbf{a}_{m-1} &= \begin{bmatrix} \mathbf{a}_{m-1,0} \\ \mathbf{a}_{m-1,1} \\ \vdots \\ \mathbf{a}_{m-1,n} \end{bmatrix}_{(n+1)d},
\end{aligned} \tag{9}$$

Box I. The structure of the matrices of (8).

3.2. Second moment

In this section we follow the steps presented in [19] to build an equation for the evolution of the second moment dynamics. We represent the second moment of (1) with the symmetric matrix

$$\bar{\bar{\mathbf{A}}}_m := \mathbb{E}[\mathbf{a}_m \mathbf{a}_m^T] = \begin{bmatrix} \bar{\bar{A}}_{m,1,1} & \bar{\bar{A}}_{m,1,2} & \cdots & \bar{\bar{A}}_{m,1,d(n+1)} \\ \bar{\bar{A}}_{m,1,2} & \bar{\bar{A}}_{m,2,2} & \cdots & \bar{\bar{A}}_{m,2,d(n+1)} \\ \vdots & \vdots & \ddots & \vdots \\ \bar{\bar{A}}_{m,1,d(n+1)} & \bar{\bar{A}}_{m,2,d(n+1)} & \cdots & \bar{\bar{A}}_{m,d(n+1),d(n+1)} \end{bmatrix}, \tag{21}$$

where

$$\bar{\bar{A}}_{m,lk} = \bar{\bar{A}}_{m,lk} := \mathbb{E}[a_{m,l} a_{m,k}] \tag{22}$$

and $a_{m,l}$ is the l th element of the vertically spread out vector $\mathbf{a}_m \in \mathbb{R}^{d(n+1)}$ defined in (9) which is given in Box I. Multiply (11) with its transpose from the right to study the evolution of the second moment:

$$\mathbf{F} \mathbf{a}_m \mathbf{a}_m^T \mathbf{F}^T = (\mathbf{F}_\tau + \mathbf{G}_\tau) \mathbf{a}_{m-1} \mathbf{a}_{m-1}^T (\mathbf{F}_\tau + \mathbf{G}_\tau)^T + (\mathbf{f} + \mathbf{g}) \mathbf{a}_{m-1}^T (\mathbf{F}_\tau + \mathbf{G}_\tau)^T + (\mathbf{F}_\tau + \mathbf{G}_\tau) \mathbf{a}_{m-1} (\mathbf{f} + \mathbf{g})^T + (\mathbf{f} + \mathbf{g})(\mathbf{f} + \mathbf{g})^T. \tag{23}$$

Applying Einstein's notation and (22) we deduce the following after taking the expected value of (23):

$$(F_{il} F_{jk}) \bar{\bar{A}}_{m,lk} = (F_{\tau,il} F_{\tau,jk}) \bar{\bar{A}}_{m-1,lk} + (F_{\tau,ik} f_j + F_{\tau,jk} f_i) \bar{\bar{a}}_{m-1,k} + (\mathbb{E}[G_{\tau,ik} g_j + G_{\tau,jk} g_i]) \bar{\bar{a}}_{m-1,k} + (f_i f_j + \mathbb{E}[g_i g_j]), \tag{24}$$

where the expected values of the products of stochastic terms are calculated using the Ito isometry [34]. Transform the system from the matrix-to-matrix mapping into a vector-to-vector representation by only including the unique elements that are in the upper-triangular diagonals of the second moment matrix (21):

$$\bar{\bar{\mathbf{a}}}_m := [\bar{\bar{A}}_{m,11}, \bar{\bar{A}}_{m,22}, \dots, \bar{\bar{A}}_{m,12}, \bar{\bar{A}}_{m,23}, \dots, \bar{\bar{A}}_{m,1,d(n+1)}]^T. \tag{25}$$

We write the evolution of the second moment dynamics in a concise form using the above defined vector representation:

$$\bar{\bar{\mathbf{F}}} \bar{\bar{\mathbf{a}}}_m = (\bar{\bar{\mathbf{F}}}_\tau + \bar{\bar{\mathbf{G}}}_\tau) \bar{\bar{\mathbf{a}}}_{m-1} + \bar{\bar{\mathbf{h}}} \bar{\bar{\mathbf{a}}}_{m-1} + (\bar{\bar{\mathbf{f}}} + \bar{\bar{\mathbf{g}}}), \tag{26}$$

where we calculate the elements of the coefficient matrices considering the index representation displayed in (24).

We reach the stationary solution of the second moment by assuming, that the first moment is already stationary $\bar{\bar{\mathbf{a}}}_m = \bar{\bar{\mathbf{a}}}_{\text{st}}$ and the present and delayed states of the squared process reach the same expected values: $\bar{\bar{\mathbf{a}}}_{\text{st}} := \bar{\bar{\mathbf{a}}}_m = \bar{\bar{\mathbf{a}}}_{m-1}$. After substituting back to (26) we conclude the following equation for the stationary second moment equation:

$$\bar{\bar{\mathbf{F}}} \bar{\bar{\mathbf{a}}}_{\text{st}} = (\bar{\bar{\mathbf{F}}}_\tau + \bar{\bar{\mathbf{G}}}_\tau) \bar{\bar{\mathbf{a}}}_{\text{st}} + \bar{\bar{\mathbf{h}}} \bar{\bar{\mathbf{a}}}_{\text{st}} + \bar{\bar{\mathbf{f}}} + \bar{\bar{\mathbf{g}}}. \tag{27}$$

Note, that the second moment is stable if and only if the first moment is stable, i.e., $\rho(H_1) < 1$ [10]. After reordering (27) we express the stationary second moment as follows:

$$\bar{\bar{\mathbf{a}}}_{\text{st}} = (\bar{\bar{\mathbf{F}}} - \bar{\bar{\mathbf{F}}}_\tau - \bar{\bar{\mathbf{G}}}_\tau)^{-1} (\bar{\bar{\mathbf{h}}} \bar{\bar{\mathbf{a}}}_{\text{st}} + \bar{\bar{\mathbf{f}}} + \bar{\bar{\mathbf{g}}}). \tag{28}$$

The second moment stability of this stationary solution is based on the mapping between the perturbed coefficients $\bar{\bar{\mathbf{a}}}_m^* = \bar{\bar{\mathbf{a}}}_m - \bar{\bar{\mathbf{a}}}_{\text{st}}$ and $\bar{\bar{\mathbf{a}}}_{m-1}^* = \bar{\bar{\mathbf{a}}}_{m-1} - \bar{\bar{\mathbf{a}}}_{\text{st}}$. After substituting back to (26) we reach the following equation:

$$\bar{\bar{\mathbf{F}}} (\bar{\bar{\mathbf{a}}}_m^* + \bar{\bar{\mathbf{a}}}_{\text{st}}) = (\bar{\bar{\mathbf{F}}}_\tau + \bar{\bar{\mathbf{G}}}_\tau) (\bar{\bar{\mathbf{a}}}_{m-1}^* + \bar{\bar{\mathbf{a}}}_{\text{st}}) + \bar{\bar{\mathbf{h}}} \bar{\bar{\mathbf{a}}}_{\text{st}} + \bar{\bar{\mathbf{f}}} + \bar{\bar{\mathbf{g}}}. \tag{29}$$

We already have an expression for the stationary solution $\bar{\bar{\mathbf{a}}}_{\text{st}}$ in (28). The stability of the stationary second moment is concluded after substituting (28) into (29):

$$\bar{\bar{\mathbf{F}}} \bar{\bar{\mathbf{a}}}_m^* = (\bar{\bar{\mathbf{F}}}_\tau + \bar{\bar{\mathbf{G}}}_\tau) \bar{\bar{\mathbf{a}}}_{m-1}^*. \tag{30}$$

From here, we describe the evolution of the second moment with the second moment mapping matrix $\mathbf{H}_2$:

$$\mathbf{H}_2 = \bar{\bar{\mathbf{F}}}^{-1} (\bar{\bar{\mathbf{F}}}_\tau + \bar{\bar{\mathbf{G}}}_\tau). \tag{31}$$

We use the spectral radius of the second moment mapping matrix $\rho(\mathbf{H}_2)$ to determine the stability of the mapping: asymptotically stable if $\rho(\mathbf{H}_2) < 1$.

4. Numerical case studies

In the previous sections, we showed how to use the stochastic collocation method to determine the stationary moment solutions and their stability for linear SDDEs. In this section, we test the convergence properties of the method through numerical case studies. Here, we focus on the computation of the second moment stability and the stationary second moment, and we do not consider the convergence of the first moment. For the first moment computation, we utilize only the deterministic component of the differential equation, and the collocation method's exponential convergence for such problems is already shown [26,27]. The numerical tests were carried out on a personal computer with an 8 core 2.9 GHz CPU and 24 GB RAM, although we focus only on the tendency of the computation time and not the absolute value.

We use the scalar stochastic Hayes equation as presented in [19], and a second order differential equation representing an engineering control problem with feedback delay as our examples. The stochastic Hayes (SH) equation is the linear scalar SDDE of the form

$$dx(t) = Ax(t)dt + [\beta x(t-1) + \gamma]dW_t, \quad (32)$$

where A and β are parameters, the delay is $\tau = 1$ and the additive noise intensity is set to $\gamma = 0.1$. For the second order (SO) equation, we consider the following SDDE

$$d\mathbf{x} = [\mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{x}(t-\tau)]dt + [\beta\mathbf{x}(t-\tau) + \gamma]dW_t, \quad (33)$$

where $\mathbf{x}$ is $\mathbb{R}^2$ -valued and

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 & 0 \\ -p & -d \end{bmatrix}, \beta = \begin{bmatrix} 0 & 0 \\ \sigma_0 p & \sigma_0 d \end{bmatrix}, \gamma = \begin{bmatrix} 0 \\ \sigma_0 \end{bmatrix}. \quad (34)$$

The system (33) describes an unstable mechanical system in a noisy environment controlled with delayed feedback control, e.g. an inverted pendulum stabilized with delayed PD control in the presence of noise. The noise intensity is $\sigma_0 = 0.1$, the delay is $\tau = 0.3$ and the parameters p and d represent the proportional and derivative gains of the PD controller [5,35].

Along with the convergence properties we also display the stability charts and stationary moments for both systems.

4.1. Convergence properties

We introduce the error measures

$$\epsilon_\rho := |\rho_2 - \rho_{2,\text{ref}}| \quad \text{and} \quad \epsilon_{\text{st}} := \left| \mathbb{E}[x_{\text{st}}^2] - \mathbb{E}[x_{\text{st}}^2]_{\text{ref}} \right| \quad (35)$$

to investigate the convergence of the spectral radius of the second moment mapping matrix (31) and the convergence of the stationary second moment (28), respectively. Here ρ_2 denotes the spectral radius of the second moment mapping matrix, i.e., $\rho_2 := \rho(\mathbf{H}_2)$. We discuss the calculation of the corresponding reference values in the next sections.

4.1.1. Spectral radius convergence

In this section, we investigate the convergence of the spectral radius ρ_2 of the second moment mapping matrix $\mathbf{H}_2$ introduced in (31). We calculate the spectral radius error ϵ_ρ and the required computational time CPU_ρ for both the SH and the SO equations with respect to the number collocation points $n+1$. For the SH equation (32) we choose $A = -8$ and $\beta = 4$, that results in $\rho_{2,\text{ref}} = 1$ as they satisfy the analytical stability boundary equation $A = -\beta^2/2$ [19]. In the case of the SO equation (33) we use a more general $p = 8$ and $d = 4$ from the stable domain. We use fourth-order stochastic semidiscretization [19] with timestep $\Delta t = \tau/100$ to obtain the reference value $\rho_{2,\text{ref}}$ because there is no analytical solution for the SO equation.

In Fig. 2 we show the relationship between the number of applied collocation points, the required computation time and the spectral radius error for the SH and SO equations.

We indicate an approximate convergence rate with dashed lines with the corresponding numerical values of the rates. Comparing the top (SO equation) and bottom rows (SH equation) of Fig. 2, we conclude that there is no qualitative difference in the behaviour of the error convergence rate and computational time of the method between the two systems. As the behaviour of the spectral radius error is similar for both test equations (SH and SO) within the reliability of the numerical tests, we summarize the findings generally, not separately for each equation.

Fig. 2(a) and (b) display the behaviour of the spectral radius error ϵ_ρ w.r.t. the number of collocation points $n+1$. It is expected, that the increase of $n+1$ decreases the spectral radius error: with a better approximation of the solution comes a more precise result, and Fig. 2(a) (b) confirm this assumption. However, there is a slight difference in the convergence rates: in case of the SO equation, the error converges with a rate of -2.5 , while in case of the SH equation the error converges with a slower rate of -2 . There are multiple potential explanations for the faster convergence rates: it can be due to the higher integration order, due to a shorter time-delay, or the numerical values of the parameters used during the analysis.

In case we consider the computational cost of the method in Fig. 2(c) and (d) we see an approximately fourth-order increase in CPU_ρ time when increasing the number of collocation points $n+1$. This mainly originates from the fact that the number of elements in the second moment mapping matrix $\mathbf{H}_2$ (31) is proportional to n^4 , and during the computation we assemble the matrix element-by-element.

Fig. 2(e) and (f) present the computational cost CPU_ρ of achieving a desired accuracy. Increasing the number of collocation points $n+1$ leads to an error $\epsilon_\rho \propto (n+1)^{-2}$, but also to a computational time cost of $\text{CPU}_\rho \propto (n+1)^4$ therefore, the method achieves a -0.5 effective convergence rate, namely $\epsilon_\rho \propto (\text{CPU}_\rho)^{-0.5}$. Depending on our computational capacity, the exact calculation time may vary, but we will observe more-or-less the same behaviour.

4.1.2. Stationary second moment convergence

In this section we investigate the error of the stationary second moment approximated using (28). For the convergence analysis, we choose $A = -8$, $\beta = 3.5$ (SH equation) and $p = 8$, $d = 4$ (SO equation). For the SH equation we use the analytical formula $\mathbb{E}[x_{\text{st}}^2] = -\sigma^2/(2A + \beta^2)$ from [19] and for the SO equation fourth-order stochastic semidiscretization with $\Delta t = \tau/100$ is utilized to acquire the reference values $\mathbb{E}[x_{\text{st}}^2]_{\text{ref}}$ for the error measure ϵ_{st} .

In Fig. 3 we present the relationship between the number of applied collocation points $n+1$, the required computation time CPU_{st} and the stationary second moment error ϵ_{st} . The top row in Fig. 3 with (a) (c) (e) corresponds to the SH equation and the bottom row with (b) (d) (f) corresponds to the SO equation. Fig. 3(a) and (b) display the change of the stationary second moment error w.r.t. the number of collocation points $n+1$.

For both test cases, the error ϵ_{st} decreases with a convergence rate -2 . However, similarly as in case of the spectral radius, the effective convergence rate of the current implementation of the method is just -0.5 as shown in Fig. 3(e) and (f). This is because increasing the number of collocation points $n+1$ results in a computational time CPU_{st} increase of fourth-order, i.e. $\text{CPU}_{\text{st}} \propto (n+1)^4$.

Comparing the results of Figs. 2 and 3, we conclude that there is no qualitative difference between the convergence of the spectral radius errors and the stationary second moment. Notably, we also observe that when handling attributes related to the second moment of stochastic systems, the collocation method no longer exhibits exponential convergence. The change in the slopes in Fig. 2(b) and (f) resembles to an exponential convergence behaviour, but it is only the result of not having accurate reference values for the SO equation.

In the following section, we present complete stability charts of the test equations and stationary second moment plots along certain parameter lines.

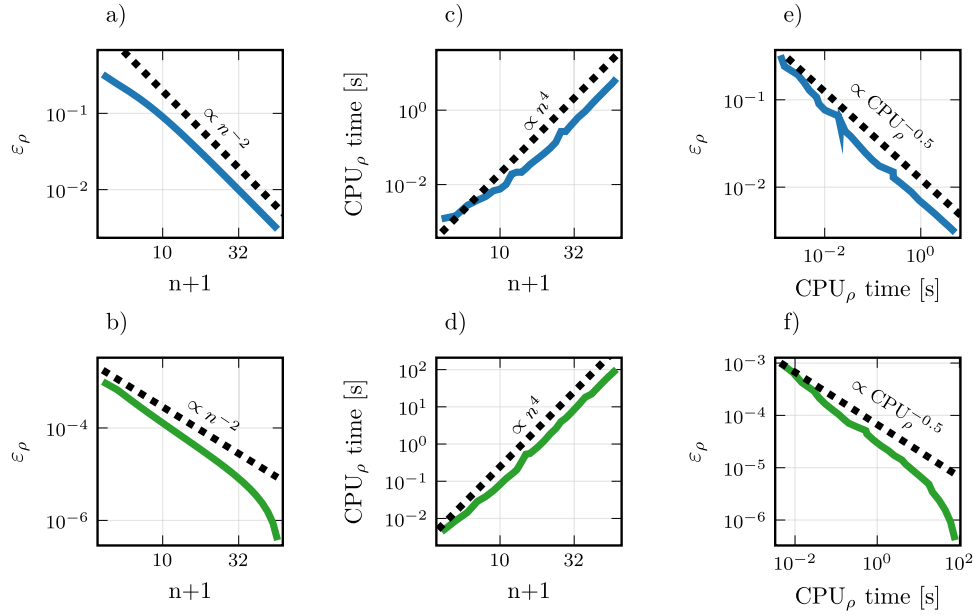


Fig. 2. The convergence properties of the spectral radius error ε_ρ of the second moment mapping matrix for the SH equation (top row) and the SO equation (bottom row). Subfigs. (a) (b) show the error and (c) (d) display the CPU_ρ time w.r.t resolution $n+1$. Subfigs. (e) (f) present the error w.r.t CPU_ρ time. In case of the SH equation we use $A = -8$, $\beta = 4$ and for the SO equation we use $p = 8$, $d = 4$. The dashed auxiliary lines are to indicate the estimated convergence rates.

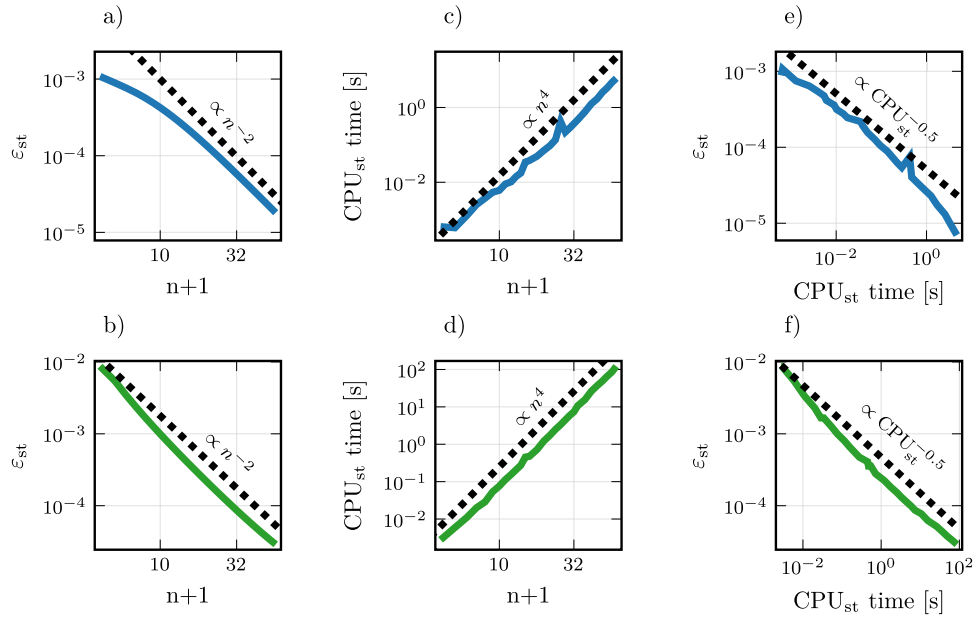


Fig. 3. The convergence properties of the stationary second moment error ε_x for the SH equation (top row) and the SO equation (bottom row). Subfigs. (a) (b) show the error and (c) (d) display the CPU_{st} time w.r.t resolution $n+1$. Subfigs. (e) (f) present the error w.r.t CPU_{st} time. In case of the SH equation we use $A = -8$, $\beta = 3.5$ and for the SO equation we use $p = 8$, $d = 4$. The dashed auxiliary lines are to indicate the estimated convergence rates.

4.2. Stationary moments and stability charts

In this section we demonstrate how the collocation method can be used to construct so-called stability charts that show the set of parameter values for which the investigated system is stable or unstable. We determine the second moment stability charts based on the spectral radius of the second moment mapping matrix (31). The borders of the stable and unstable regions are determined by the critical parameter values for which $\rho(\mathbf{H}_2) = 1$. To find these parameter values we use a multi-dimensional bisection method [36,37] in the parameter plane.

Fig. 4(a) and (c) show the borders of the stable parameter regions, while (b) and (d) show the stationary second moment for a set of chosen

parameters. The different coloured lines correspond to different number of collocation points $n+1$, while the black lines correspond to lines obtained with reference values. In case of the SH equation (32) we use $A = -\beta^2/2$ as the reference, while for the stationary second moment we use $\mathbb{E}[x_{st}^2] = -\sigma^2/(2A + \beta^2)$ from [19]. As for the SO equation (33) there is no analytical solution available, we use fourth-order semidiscretization with timestep $\Delta t = \tau/100$ to obtain the reference values. In case of the SH equation we show the stationary second moment along the parameter line $A = -6$ and vary the parameter β in Fig. 4(b), while in case of the SO equation we show the stationary second moment along the parameter line $p = 7$ and vary the parameter d in Fig. 4(d).

In the stability charts (Fig. 4(a) and (c)) we observe that the second moment stability boundary obtained using the stochastic collocation

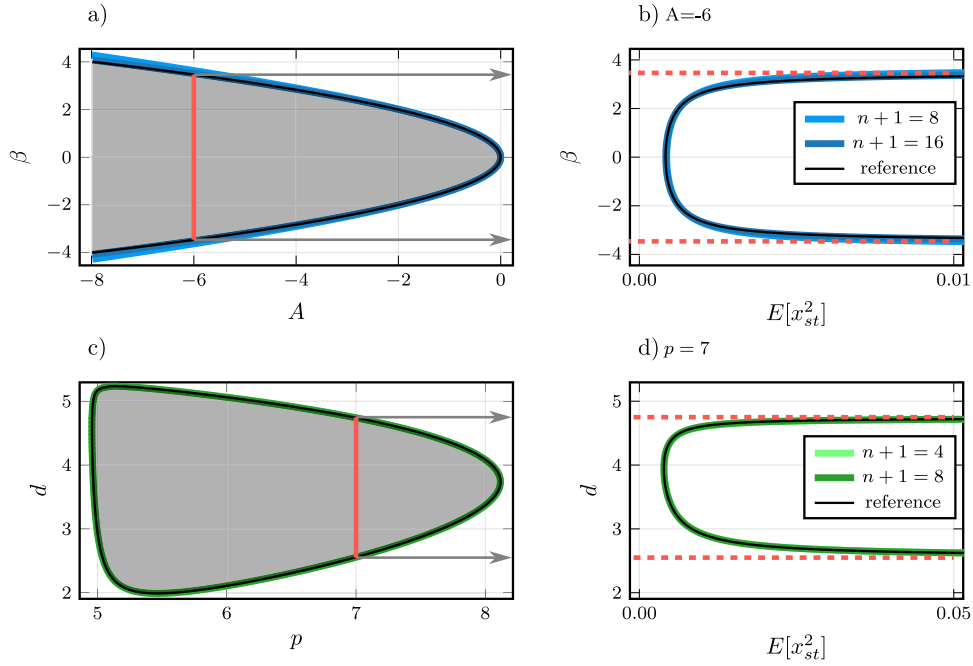


Fig. 4. SH equation: (a) the second moment stability boundary w.r.t. $n+1$ and (b) the stationary second moment in the stable region along $A = -6$. SO equation: (c) the second moment stability boundary w.r.t. $n+1$ and (d) the stationary second moment in the stable region along $p = 7$. The reference boundaries are determined by (a) (b) analytical formulas (c) (d) fourth-order stochastic semidiscretization. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

method is much more accurate for the SO equation (33) than for the SH equation (32). This can be attributed to the differences in the structures of the two systems, and to the shorter delay in the SO equation. However, we conclude, that the stability boundaries obtained with the method capture the dynamic structure of the system with high fidelity even for low number ($n+1$) of collocation points.

Furthermore, we observe in Fig. 4(b) and (d) that near the stability boundaries, the stationary second moment $\mathbb{E}[x_{st}^2]$ starts to increase exponentially towards infinity for both systems, this is called: stochastic resonance. These large stationary second moment values indicate that the realizations show large deviations from the average trajectories. The physical interpretation is that during e.g. in a control task we can stabilize the system, however, we have fluctuations about the equilibrium position. If the system and our control parameters are such, that the process operates near the boundary of the stable region, these fluctuations can become so significant, that the linear model used to investigate the system does not hold anymore. At this point, the stable second moment prediction of the linear model might fail due to the underlying non-linear dynamics.

5. Conclusion

In this paper we presented a novel, collocation based approach to approximate the second moment stability and stationary second moment of linear stochastic delay differential equations. First, we construct the first and second moment mappings between time intervals with a length equal to the time-delay utilizing Chebyshev polynomials. We analyse the convergence properties through numerical experiments using two numerical case studies: the scalar stochastic Hayes equation and a second order equation. These tests show, that as we increase the number of interpolation points $n+1$, the spectral radius and the steady state converges to their accurate solution with at least an order 2. However, the number collocation points $n+1$ also increases the size of the mapping matrices $\mathbf{H}_1 \in \mathbb{R}^{(n+1) \times (n+1)}$ and $\mathbf{H}_2 \in \mathbb{R}^{k \times k}$, where $k = (n+1)(n+2)/2$. As in the current implementation we compute each element of the second moment mapping matrix $\mathbf{H}_2$ individually, this leads to an effective second moment error convergence rate of -0.5 .

Practically, the same convergence rates are presented in case of the stochastic semidiscretization [19] as well. We can only compare the two methods qualitatively due to the different implementations and testing environments, but it appears that both methods perform similarly in terms of their second moment convergence properties. However, there could be specific applications, where one or the other proves to be more suitable.

Our future plans to improve the stochastic collocation method include an efficient way to compute the mapping matrices, e.g. optimizing the code and memory allocations, but it is beyond of the scope of this paper. In its current state, this stochastic collocation method is best to be applied for the stability and steady-state analysis of small-scale stochastic dynamical systems with delays, such as control systems in noisy environments. Furthermore, the current formulation of the method generalizes well for multiple delays and noise sources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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